Parrot fish crunches up coral all day. The source of the parrotfish's powerful bite is the interwoven fibre nanostructure.
Crystals of a mineral called fluorapatite are woven together in a chain mail-like arrangement.
This structure gives parrotfish teeth incredible durability.

Mimic in laboratories

Water repellent Nano paints are made. Coating with such nano paints give durability, protection against stains and dirt also enhances fuel efficiency when coated on ships.



Nano in Nature

Object

Lotus leaf surface Scanning electron micrograph (SEM) showing the nano structures on the surface of a leaf from a lotus plant. This is the reason for self cleaning process in lotus leaf.



Nano in Nature

Mimic in laboratories

The natural structure provides a blueprint for creating ultra-durable synthetic materials that could be useful for mechanical components in electronics, and in other devices that undergo repetitive movement, abrasion, and contact stress.

XII_Physics_Vol-2_UNIT-11_EM.indd 257 26-12-2022 15:49:16